THE MOLE 1

Many everyday names represent a fixed number of objects. For example:

- a soccer team: 11
- a netball team: 7
- a hockey team: 11
- an Australian Rules football team: 18
- a dozen eggs: 12
- a gross of woodscrews: 144
- a ream of paper: 500 sheets
- a century of runs: 100

1 Give another name that stands for a fixed quantity.

Chemistry is all about objects like atoms, molecules and ions. These objects are all so small that there must be many billions of them before they can be seen. A special number used in chemistry is 6.02×10^{23}, called the **mole**. It is also called **Avogadro's number**, in honour of Amedeo Avogadro (1776–1856). The mole is simply a convenient unit of measurement that chemists use because it fits in with 'natural' numbers. It is like saying there are 365 days in a year, the diameter of the Earth is 12 800 km, light travels at 300 000 kilometres per second (approximately), and so on.

Units of measurement are often used in mathematics questions:

EXAMPLES

i How many dozens (units of 12) are there in 36 units?

Divide the total number by the number in one dozen: $36 \div 12 = 3$ dozen.

ii How many days are there in 100 hours?

$100 \div 24 = 4.17$ days

The same mathematic principles apply to calculations with moles.

EXAMPLES

iii How many atoms of sodium are there in 0.5 moles of sodium?

'Of' is a hint to use multiplication. Multiply the number of moles by the number of atoms in a mole:
$0.5 \times 6.02 \times 10^{23} = 3.01 \times 10^{23}$

iv How many moles of water molecules are in 57×10^{97} particles of water?

Divide the total number of particles by the number of particles that makes up a mole:
$57 \times 10^{97} \div 6.02 \times 10^{23} = 9.5 \times 10^{74}$

(Note that when using moles we sometimes get 'silly' or 'ridiculously large' numbers.)

2 Calculate how many moles are present in each of the following.

a 88×10^{43} copper atoms **b** 12×10^{13} helium atoms

Masses of chemicals

Atomic mass units (amu) are used to describe the mass of atoms. Today, atomic mass units are defined in terms of the ^{12}C isotope, which is assigned a mass of exactly 12.00 amu. The amu is too small a unit to use for everyday objects: 1 amu is equivalent to 1.67×10^{-27} grams.

3 Explain what is meant by 'the isotope ^{12}C'.

Chemists need to know what mass of an element will give one mole of atoms of that element. The mole can be defined as the number of atoms in 12.00 grams of the ^{12}C isotope. That is, in 12.00 grams of the ^{12}C isotope there are 6.0221367×10^{23} atoms (commonly abbreviated to 6.02×10^{23}).

In a Periodic Table, carbon is assigned a slightly different mass: 12.01. This is because naturally occurring carbon is a mixture of isotopes; 12.01 is an average of the masses of the isotopes according to the proportion in which they occur in nature. The Periodic Table quotes hydrogen as having a mass of 1.008. This means that atoms of naturally occurring hydrogen have an average mass of 1.008 amu; therefore a mole of hydrogen atoms has a mass of 1.008 g. Helium is quoted as having a mass of 4.003. This means that atoms of naturally occurring helium have an average mass of 4.003 amu; therefore a mole of helium atoms has a mass of 4.003 g.

A **mole** of atoms of any element = the **atomic mass** of that element in grams.

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4 Give the chemical symbol, atomic mass and the mass of a mole of the following atoms:

a oxygen b neon c sodium d copper

Formula mass

The atomic mass of a molecule or compound is not referred to in chemistry because more than one atom is involved and each atom making up the molecule has its own atomic mass. Instead, the **formula mass** is used because each molecule has a specific formula. The formula mass is calculated by adding the atomic masses of each element in the molecule.

EXAMPLE

v **What is the mass of one mole of bromine?**

Bromine exists as a diatomic molecule, Br_2, where two bromine atoms make up one bromine molecule. Each bromine atom has a mass of 80.00 amu. Therefore, the formula mass of a bromine molecule is 160.00 (2 × 80.00) and one mole of bromine molecules has a mass of 160.00 g.

5 Calculate the masses of one mole of the following diatomic molecules:

a oxygen (O_2) [O = 16.00 amu]
b nitrogen (N_2) [N = 14.01 amu]
c chlorine (Cl_2) [Cl = 35.45 amu]

The same principle can be applied to work out the masses of molecules.

EXAMPLES

vi **What is the mass of a mole of magnesium chloride?**

In magnesium chloride ($MgCl_2$) there is one magnesium atom (24.31) and two chlorine atoms (35.45 × 2 = 70.90) present in the formula.

Hence the formula mass of magnesium chloride is 24.31 + 70.90 = 95.21. This is the total of all of the atomic masses of the atoms that are shown in the formula of the substance. Therefore the mass of a mole of magnesium chloride is 95.21 g.

vii **What is the mass of a mole of water?**

Water has the formula H_2O. Each molecule contains two hydrogen atoms and one oxygen atom. Hydrogen atoms have a mass of 1.008 amu and oxygen atoms a mass of 16.00 amu.

The formula mass is (2 × 1.008) + (1 × 16.00) = 18.02 amu. Therefore a mole of water molecules has a mass of 18.02 g.

6 Calculate the mass of one mole of each compound. Refer to a Periodic Table for the relevant atomic masses.

a carbon dioxide, CO_2
b phosphorus pentachloride, PCl_5
c aluminium oxide, Al_2O_3
d glucose, $C_6H_{12}O_6$
e sulfuric acid, H_2SO_4
f calcium hydroxide, $Ca(OH)_2$
g ethanoic acid, CH_3COOH

Often parts or multiples of moles are used in chemistry. To calculate these, work out the molar mass of the substance and then multiply it by the number of moles.

EXAMPLE

viii **What is the mass of 0.75 moles of water?**

The mass is the number of moles multiplied by molar mass: 0.75 × 18.02 = 13.52 g

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7 For each of the following, give the formula and calculate the mass in grams.

a 0.5 moles of carbon dioxide
b 0.1 moles of hydrogen chloride
c 2 moles of sulfuric acid
d 0.1 moles of phosphorus pentachloride
e 0.25 moles of magnesium nitrate
f 1.5 moles of ammonia
g 1 mole of chlorine molecules

Chemists need to work out how many moles of a substance are present in a given mass. To do this, calculate the mass of one mole (atomic, molecular or formula mass in grams). Then divide the given mass by the mass of one mole.

$$\text{number of moles} = \frac{\text{given mass}}{\text{molar mass}}$$

EXAMPLE

ix **How many moles of ammonia molecules are present in 30.00 g of ammonia gas?**

Ammonia has the formula NH_3. Calculate its formula mass:
$(1 \times N) + (3 \times H) = (1 \times 14.01) + (3 \times 1.008) = 17.03$

One mole of ammonia has a mass of 17.03 g or 17.03 g of ammonia is one mole.
Divide the given mass by 17.03 to calculate the number of moles of ammonia: $30.00 \div 17.03 = 1.762$.
Given that there are 30.00 g of the substance and 17.03 g per mole, calculate the number of moles: $30.00 \div 17.03 = 1.762$. Therefore there are **1.762 moles** of ammonia in 30.00 g of ammonia gas.

8 Calculate how many moles are in each mass.

a 12.0 g of helium atoms
b 20.0 g of bromine atoms
c 35.5 g of chlorine molecules
d 20.0 g of sodium hydroxide
e 200 g of calcium carbonate
f 5.60 g of iron
g 9.80 g of sulfuric acid

Equations and masses

Simple reaction equations can be deduced from relative masses of pure substances. First, find the number of moles present of each substance (divide mass present by formula mass) and then work out the ratio of moles of each substance.

EXAMPLE

x **7.95 g of copper (II) oxide reacts with 0.60 g of carbon to produce 6.35 g of copper and 2.2 g of carbon dioxide. Work out the equation for the reaction.**

copper (II) oxide (CuO): formula mass = 63.55 + 16.00 = 79.55 amu
carbon (C): atomic mass = 12.01 amu
carbon dioxide (CO_2): formula mass = 12.01 + (2 × 16.00) = 12.01 + 32.00 = 44.01 amu
No. of moles of copper oxide = 7.95 ÷ 79.55 = 0.1 moles
No. of moles of carbon = 0.60 ÷ 12.01 = 0.05 moles
No. of moles of carbon dioxide = 2.2 ÷ 44.01 = 0.05 moles
The ratio CuO : C : CO_2 = 0.1 : 0.05 : 0.05 = 2 : 1 : 1
That is, the reaction equation is **$2CuO\ (s) + C\ (s) \rightarrow 2Cu\ (s) + CO_2\ (g)$**

9 Work out the equation for the reaction between 40.00 g of iron (III) oxide and 21.00 g of carbon monoxide, producing 28.00 g of iron and 33.00 g of carbon dioxide.

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Empirical formulae

The empirical formula of a compound is the simplest formula that shows the ratio of the different atoms in the compound.

The empirical formula for a compound is the simplest ratio of atoms present. The molecular formula shows the actual number of atoms present. For some compounds the empirical and molecular formulae are the same. The empirical formula of hydrogen peroxide is HO: there is one hydrogen atom for every oxygen atom. The molecular formula of hydrogen peroxide is actually H_2O_2: there are two atoms of hydrogen and two atoms of oxygen in every molecule of hydrogen peroxide.

10 Write the empirical formula for each substance.

- **a** glucose, $C_6H_{12}O_6$
- **b** butane, C_4H_{10}
- **c** ethanoic acid, CH_3COOH
- **d** aluminium oxide, Al_2O_3

The empirical formula and the molecular mass can be used to find the molecular formula of some substances.

EXAMPLE

xi A hydrocarbon has an empirical formula of CH_2. It has a formula mass of 84.00. What is its molecular formula?

There are two hydrogens for every carbon atom, which means the molecular formula is a multiple of CH_2, such as C_2H_4 or C_3H_6. This can be written as $(CH_2)_n$.

CH_2 is one carbon atom (1×12.01) plus two hydrogen atoms (2×1.008) = 12.01 + 2.016 = 14.02.

To find the number of CH_2 units, divide the formula mass by the mass of the empirical formula: 84.00 ÷ 14.02 = 6. This means that there are 6 groups of CH_2 in a molecule of the hydrocarbon, so the molecular formula is C_6H_{12}.

11 A hydrocarbon has twice as many hydrogen atoms as carbon atoms. Its relative molecular mass is approximately 112 amu.

- **a** Calculate its empirical formula.
- **b** Calculate its molecular formula.

The relative masses of the elements in a compound depend on the masses and the relative numbers of the atoms present. If the total mass of a compound and the individual masses of the elements that make it up are known, the relative numbers of atoms of each element present can be worked out. Divide the masses of elements present by their relative atomic masses (ram). Then find the ratio of these numbers. In other words, use the number of moles of different atoms present in a substance to work out the empirical formula of that substance.

EXAMPLE

xii 11.00 g of a compound of mercury and chlorine contained 9.34 g of mercury. The relative molecular mass (rmm) was found to be 472.00 amu.

- **Calculate the empirical formula of the compound.**
- **Calculate the molecular formula of the compound.**

– Calculate the mass of the reactants: mercury = 9.34 g; chlorine = (11.00 – 9.34) = 1.66 g

– Calculate the number of moles:
Mercury (Hg) has a ram of 200.6 amu. No. of moles = 9.34 ÷ 200.6 = 0.046 moles.
Chlorine (Cl) has a ram of 35.50 amu. No. of moles = 1.66 ÷ 35.50 = 0.046 moles

– Calculate the ratio. 0.046 : 0.046 or 1 : 1
Hence the empirical formula is HgCl.

– Use the empirical formula to calculate rmm: 200.60 + 35.45 = 236.05 amu

– Work out the molecular formula:
Actual rmm = 472.00 amu. 472.00 ÷ 236.10 = 2. Hence the molecular formula is Hg_2Cl_2.

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xiii 1.32 g of magnesium was dissolved in dilute hydrochloric acid. The solution was then heated in a stream of hydrogen chloride. 5.26 g of the anhydrous metal chloride was formed. Find the simplest formula for the metal chloride.

1.32 g of magnesium combined with (5.26 – 1.32) g of chlorine = 3.94 g of chlorine

Moles of Mg atoms : moles of Cl atoms
= (1.32 ÷ 24) : (3.94 ÷ 35.5)
= 0.055 : 0.111 = 1 : 2 (to the nearest whole number)

Therefore the simplest formula is $MgCl_2$.

12 Two oxides of nitrogen were analysed. Both oxides contain two atoms of nitrogen per molecule. The first oxide has a rmm of 76 amu and the other oxide has a rmm of 108 amu. Calculate the molecular formula of each compound. Show your working.
(Hint: work out how much nitrogen is present per molecule, the remainder must be oxygen.)

13 0.48 g of magnesium is burned in oxygen to form 0.80 g of magnesium oxide.

a How much oxygen is used in the combustion?

b What is the empirical formula of magnesium oxide? Show your working.

Percentages and empirical formula

If the percentages (by mass) of elements in a substance are known, the empirical formula of the substance can be calculated.

EXAMPLE

xvi A chemical analysis of an organic compound showed that it was made up of carbon (40%), hydrogen (6.7%) and oxygen (53.3%). What is its empirical formula?

Elements present:	C	H	O
Percentage:	40	6.6	53.3
ram:	12.01	1.008	16.00
percentage ÷ ram	40 ÷ 12.01	6.6 ÷ 1.008	53.3 ÷ 16.00
	= 3.33	= 6.66	= 3.33
divide by lowest common denominator (3.33)	1	2	1

Hence the empirical formula is CH_2O.

14 A compound has an empirical formula CH_2O. Its relative molecular mass (rmm) is 180.20 amu. Work out its molecular formula.

15 Car radiators often contain antifreeze. One particular type of antifreeze was analysed and found to contain carbon (38.7%), hydrogen (9.7%) and oxygen (51.6%).

a Calculate its empirical formula.

b Given that its rmm is 62.10 amu, what is its molecular formula?

16 Hydrated copper sulfate crystals were analysed and gave this percentage composition by mass: Cu = 25.8%, S = 12.8%, O = 25.6% and water of crystallisation = 36.1%. What is the empirical formula of the crystals? (Hint: treat the water of crystallisation by itself.)

17 Mark each statement as true or false.

a A mole of particles consists of Avogadro's number.

b The empirical formula of a substance shows the actual numbers and types of atom present.

c Relative molecular mass (rmm) is measured in atomic mass units (amu).

d The term 'mole' comes from the abbreviation for Molekulargewicht (molecular weight).

e Avogadro's number was discovered by Amedeo Avogadro.

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